

'-- sql injection

the old dog of data security



LIVE

BREAKING

NEWS

CHANNEL 10

URGENT

**BREAKING
NEWS**

BREAKING NEWS

Another embarrassing data breach. Personal data all over the Internet.

• No clue how this happened: CIO • It was a ticking time bomb all this time: DBA •

'-- the evergreen vulnerability

And although declining in recent years, it just won't go away.

- 6.7% of all vulnerabilities in open-source.
- 10% of all vulnerabilities in closed-source. (2024)
- Ranked #5 vulnerability in 2025 (OWASP), down from #3.



'-- in the news

You may have heard about some of these fine companies?

- Sony Playstation Network
- Equifax
- Yahoo
- Epic Games / Fortnite
- SQL Server Reporting Services
- 7-Eleven
- Sony Pictures
- Marriott International
- TSA



The good news is, it's relatively easy to fix.



learning objectives

- What is a SQL injection
- What are the risks
- What to look for
- How to fix your code



so, who am I?

'-- daniel hutmacher



sql server consultant
conference organizer
dog person

sqlsunday.com
daniel@strd.co



@dhma.ch



/in/danielhutmacher/





I talked to some sponsors in the break



Be honest



I am being honest



What did you talk about?



I picked up some cool swag



Thank you.

twintos



meet Ali



Ali

Sales rep of the year, 2024

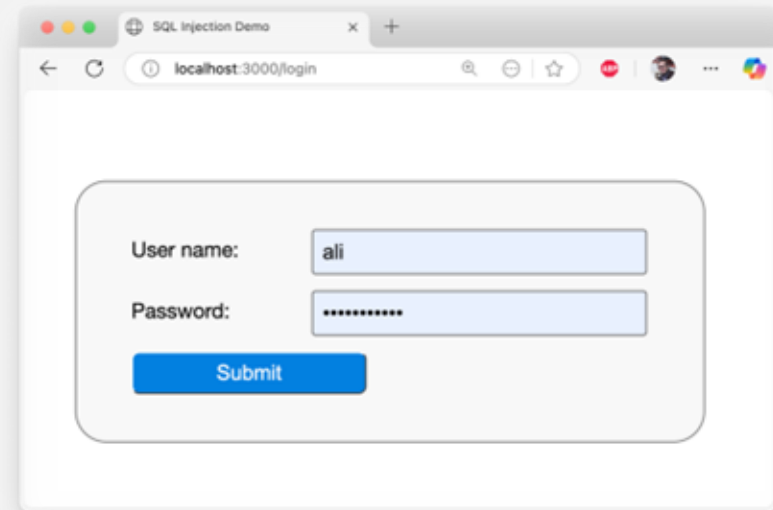
Good boy

meet Ali



Ali

Sales rep of the year, 2024
Good boy



meet Ali



Ali

*Sales rep of the year, 2024
Good boy*

A screenshot of a web browser window titled "SQL Injection Demo". The address bar shows "localhost:3000/login". The login form has two input fields: "User name:" with the value "ali" and "Password:" with masked characters "*****". Below the fields is a blue "Submit" button.

http

POST /login

username=ali

password=password123

meet Ali



Ali

*Sales rep of the year, 2024
Good boy*

A screenshot of a web browser window titled "SQL Injection Demo". The address bar shows "localhost:3000/login". The login form has two input fields: "User name:" with the value "ali" and "Password:" with masked characters "*****". A blue "Submit" button is below the fields.

http

```
POST /login
username=ali
password=password123
```



sql

```
SELECT UserName
FROM dbo.SalesAgents
WHERE UserName='ali'
AND PasswordText='password123';
```

what is sql injection?

powershell

```
$email = "daniel@strd.co"  
  
$sqlQuery = @"  
    SELECT id, firstName, lastName  
    FROM dbo.Users  
    WHERE email='$email';  
"@
```



sql

```
SELECT id, firstName, lastName  
FROM dbo.Users  
WHERE email='daniel@strd.co';
```



what is sql injection?

powershell

```
$email = "daniel@strd.co' OR 1=1 --"  
$sqlQuery = @"  
    SELECT id, firstName, lastName  
    FROM dbo.Users  
    WHERE email='$email';  
"@
```



sql

```
SELECT id, firstName, lastName  
FROM dbo.Users  
WHERE email='daniel@strd.co' OR 1=1 --';
```



types of sql injection



types of sql injection

out-of-band attack

powershell

```
$email = "daniel@strd.co";  
CREATE LOGIN legit WITH PASSWORD='1234';  
GRANT CONTROL SERVER TO legit; --"
```

```
$sqlQuery = @"  
SELECT id, firstName, lastName  
FROM dbo.Users  
WHERE email='$email';  
"@
```



sql

```
SELECT id, firstName, lastName  
FROM dbo.Users  
WHERE email='daniel@strd.co';  
CREATE LOGIN legit WITH PASSWORD='1234';  
GRANT CONTROL SERVER TO legit; --';
```



types of sql injection

out-of-band attack

powershell

```
$email = "daniel@strd.co";  
xp_cmdshell 'del C:\Windows\System32\*'; --"  
  
$sqlQuery = @"  
    SELECT id, firstName, lastName  
    FROM dbo.Users  
    WHERE email='$email';  
"@
```



sql

```
SELECT id, firstName, lastName  
FROM dbo.Users  
WHERE email='daniel@strd.co';  
xp_cmdshell 'del C:\Windows\System32\*'; --';
```



types of sql injection

error-based injection

powershell

```
$email = "daniel@strd.co' OR 1=CAST(@@SERVERNAME AS int) --"  
$sqlQuery = @"  
    SELECT id, firstName, lastName  
    FROM dbo.Users  
    WHERE email='$email';  
"@
```



sql

```
SELECT id, firstName, lastName  
FROM dbo.Users  
WHERE email='daniel@strd.co' OR 1=CAST(@@SERVERNAME AS int) --';
```



types of sql injection

error-based injection

powershell

```
$email = "daniel@strd.co' OR 1=CAST(@@SERVERNAME AS int) --"  
$sqlQuery = @"  
    SELECT id, firstName, lastName  
    FROM dbo.Users  
    WHERE email='$email';  
"@
```



sql

```
SELECT id, firstName, lastName  
FROM dbo.Users  
WHERE email='daniel@strd.co' OR 1=CAST(@@SERVERNAME AS int) --';
```

Msg 245, Level 16, State 1, Line 1
Conversion failed when converting the nvarchar value 'SQL01' to data type int.



types of sql injection

error-based injection

-- demo



Test-Injection.ps1



types of sql injection

blind injection

powershell

```
$email = "daniel@strd.co"; IF (SUSER_NAME()='sa') WAITFOR DELAY '00:00:01'; --"  
$sqlQuery = @"  
    SELECT id, firstName, lastName  
    FROM dbo.Users  
    WHERE email='$email';  
"@
```



sql

```
SELECT id, firstName, lastName  
FROM dbo.Users  
WHERE email='daniel@strd.co'; IF (SUSER_NAME()='sa') WAITFOR DELAY '00:00:01'; --';
```



types of sql injection

blind injection

sqlmap demo



```
Daniels zsh
~/Desktop/sqlmap 🐚 python3 sqlmap.py -u http://localhost:3000/login --data="username=hello&password=helloagain" --tables
--columns -dbms=MSSQL -D InjectionDemo
```



let's inject some sales

because we're worth it.



-- demo



Injection demo code.txt

dynamic sql

ETL frameworks can be vulnerable, too.

ID	Source_object	Dest_object	Dependency	TimestampColumn
1	Staging.Currencies	dbo.Currencies	NULL	Updated
2	Staging.CurrencyRates	dbo.Fx_Rates	1	Updated
3	Staging.Contracts	dbo.Contracts	2	Updated



dynamic sql

ETL frameworks can be vulnerable, too.



ID	Source_object	Dest_object	Dependency	TimestampColumn
1	Staging.Currencies	dbo.Currencies	NULL	Updated
2	Staging.CurrencyRates	dbo.Fx_Rates	1	Updated
3	Staging.Contracts	dbo.Contracts	2	Updated

sql

```
SELECT TOP (1) @sql='
  INSERT INTO '+Dest_object+
  SELECT * FROM '+Source_object+
  WHERE '+TimestampColumn+'>' +
  '(SELECT MAX('+TimestampColumn+') FROM '+Dest_object+');'
FROM Metadata.ETL_flows;

EXEC(@sql);
```

dynamic sql

ETL frameworks can be vulnerable, too.



ID	Source_object	Dest_object	Dependency	TimestampColumn
1	Staging.Currencies	dbo.Currencies	NULL	Updated
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sql

```
SELECT TOP (1) @sql='
  INSERT INTO '+Dest_object+
  SELECT * FROM '+Source_object+
  WHERE '+TimestampColumn+'>'+
    '(SELECT MAX('+TimestampColumn+') FROM '+Dest_object+');'
FROM Metadata.ETL_flows;

EXEC(@sql);
```



sql

```
INSERT INTO dbo.Currencies
SELECT * FROM Staging.Currencies
WHERE Updated>(SELECT MAX(Updated) FROM dbo.Currencies);
```

dynamic sql

ETL frameworks can be vulnerable, too.



ID	Source_object	Dest_object	Dependency	TimestampColumn
1	Staging.Currencies	dbo.Currencies	NULL	1=0; DROP TABLE dbo.AuditLog; UPDATE Metadata.ETL_flows SET TimestampColumn='Updated'--
2	Staging.CurrencyRates	dbo.Fx_Rates	1	Updated
3	Staging.Contracts	dbo.Contracts	2	Updated



lps

```
INSERT INTO dbo.Currencies
SELECT * FROM Staging.Currencies
WHERE 1=0; DROP TABLE dbo.AuditLog; UPDATE Metadata.ETL_flows SET
TimestampColumn='Updated'-->(SELECT MAX(1=0; DROP TABLE dbo.AuditLog; UPDATE
Metadata.ETL_flows SET TimestampColumn="Updated"--
) FROM dbo.Currencies);
```

dynamic sql

ETL frameworks can be vulnerable, too.



ID	Source_object	Dest_object	Dependency	TimestampColumn
1	Staging.Currencies	dbo.Currencies	NULL	1=0; DROP TABLE dbo.AuditLog; UPDATE Metadata.ETL_flows SET TimestampColumn='Updated'--
2	Staging.CurrencyRates	dbo.Fx_Rates	1	Updated
3	Staging.Contracts	dbo.Contracts	2	Updated



1. finish the INSERT:

lps

```
INSERT INTO dbo.Currencies  
SELECT * FROM Staging.Currencies  
WHERE 1=0;
```

```
DROP TABLE dbo.AuditLog;
```

```
UPDATE Metadata.ETL_flows SET TimestampColumn='Updated'-->(SELECT MAX(1=0; DROP  
TABLE dbo.AuditLog; UPDATE Metadata.ETL_flows SET TimestampColumn="Updated"--  
) FROM dbo.Currencies);
```


dynamic sql

ETL frameworks can be vulnerable, too.



ID	Source_object	Dest_object	Dependency	TimestampColumn
1	Staging.Currencies	dbo.Currencies	NULL	1=0; DROP TABLE dbo.AuditLog; UPDATE Metadata.ETL_flows SET TimestampColumn='Updated'--
2	Staging.CurrencyRates	dbo.Fx_Rates	1	Updated
3	Staging.Contracts	dbo.Contracts	2	Updated



1. finish the INSERT:

2. do bad things:

lps

```
INSERT INTO dbo.Currencies
SELECT * FROM Staging.Currencies
WHERE 1=0;
```

```
DROP TABLE dbo.AuditLog;
```

```
UPDATE Metadata.ETL_flows SET TimestampColumn='Updated'-->(SELECT MAX(1=0; DROP
TABLE dbo.AuditLog; UPDATE Metadata.ETL_flows SET TimestampColumn="Updated"--
) FROM dbo.Currencies);
```

dynamic sql

ETL frameworks can be vulnerable, too.



ID	Source_object	Dest_object	Dependency	TimestampColumn
1	Staging.Currencies	dbo.Currencies	NULL	1=0; DROP TABLE dbo.AuditLog; UPDATE Metadata.ETL_flows SET TimestampColumn='Updated'--
2	Staging.CurrencyRates	dbo.Fx_Rates	1	Updated
3	Staging.Contracts	dbo.Contracts	2	Updated



1. finish the INSERT:

2. do bad things:

3. remove evidence:

sql

```
INSERT INTO dbo.Currencies
SELECT * FROM Staging.Currencies
WHERE 1=0;
```

```
DROP TABLE dbo.AuditLog;
```

```
UPDATE Metadata.ETL_flows SET TimestampColumn='Updated'-->(SELECT MAX(1=0; DROP
TABLE dbo.AuditLog; UPDATE Metadata.ETL_flows SET TimestampColumn="Updated"--
) FROM dbo.Currencies);
```



but wait, there's more:

connection string injection

connection string injection



powershell

```
$username = "daniel"  
$password = "Pa$$w0rd!"  
  
$connectionString = "Server=sql01.strd.co;Database=Prod_DB;User  
Id=$username;Password=$password;"
```



connection string

```
Server=sql01.strd.co;Database=Prod_DB;User Id=daniel;Password=Pa$$w0rd!;
```

connection string injection



powershell

```
$username = "daniel"  
$password = "Pa$$w0rd!;Integrated Security=True"  
  
$connectionString = "Server=sql01.strd.co;Database=Prod_DB;User  
Id=$username;Password=$password;"
```



connection string

```
Server=sql01.strd.co;Database=Prod_DB;User Id=daniel;Password=Pa$$w0rd!;Integrated  
Security=True;
```

'-- mitigations

- Parameterizing inputs
- Manually sanitizing inputs
- Using an ORM
- Using stored procedures
- Restrictive permissions
- Disabling error messages on web sites
- Web application firewalls (WAF)
- Rate-limit your web app or API



mitigations



parameterizing inputs

#6

```
string query = "SELECT id, firstName, lastName FROM dbo.Users WHERE email = @Email";  
SqlCommand command = new SqlCommand(query, connection);  
  
command.Parameters.Add(  
    new SqlParameter("@Email", SqlDbType.NVarChar, 255) {  
        Value = email  
    }  
);
```



mitigations

parameterizing dynamic t-sql



lps

```
SET @sql=N'  
  UPDATE dbo.Users  
  SET firstName=@first, lastName=@last  
  WHERE email=@email;'  
  
EXECUTE sys.sp_executesql  
  @sql,  
  @params = N'@first nvarchar(100), @last nvarchar(100), @email varchar(255)',  
  @first = @new_firstname,  
  @last = @new_lastname,  
  @email = @email;
```



mitigations



use an ORM

using parameterized SQL
(EF Core 2.0+)

C#

```
var email = "daniel@strd.co";  
  
var users = context.Users  
    .FromSqlInterpolated($"SELECT id, firstName, lastName FROM dbo.Users WHERE email =  
    {email}")  
    .ToList();
```

using DbSet.SqlQuery
(EF6+)

C#

```
var email = "daniel@strd.co";  
  
var users = context.Users  
    .SqlQuery("SELECT id, firstName, lastName FROM dbo.Users WHERE email = @p0", email)  
    .ToList();
```



mitigations

sanitizing dynamic sql



ID	Source_object	Dest_object	Dependency	TimestampColumn
1	Staging.Currencies	dbo.Currencies	NULL	Updated
2	Staging.CurrencyRates	dbo.Fx_Rates	1	Updated
3	Staging.Contracts	dbo.Contracts	2	Updated

sql

```
SELECT TOP (1) @sql='  
  INSERT INTO '+Dest_object+'  
  SELECT * FROM '+Source_object+'  
  WHERE '+QUOTENAME(TimestampColumn)+'>'+  
    '(SELECT MAX('+QUOTENAME(TimestampColumn)+') FROM '+Dest_object+');'  
FROM Metadata.ETL_flows;  
  
EXEC(@sql);
```



sql

```
INSERT INTO dbo.Currencies  
SELECT * FROM Staging.Currencies  
WHERE [Updated]>(SELECT MAX([Updated]) FROM dbo.Currencies);
```



mitigations

sanitizing dynamic sql

I ❤️ QUOTENAME()

sql

```
SELECT QUOTENAME('hello');  
SELECT QUOTENAME('object', '[');  
SELECT QUOTENAME('quote', '"');  
SELECT QUOTENAME('apostrophe', "'");
```



```
[hello]  
[object]  
"quote"  
'apostrophe'
```



mitigations

sanitizing dynamic sql

- OBJECT_ID() returns the unique id of a table/view/etc.
- OBJECT_NAME() returns the name of an object.
- OBJECT_SCHEMA_NAME returns the name of the schema of an object.
- QUOTENAME quotes (and sanitizes) a string.

sql

```
-- DECLARE @Source_object nvarchar(100)='Staging.Currencies';  
  
SELECT  
    QUOTENAME(OBJECT_SCHEMA_NAME(OBJECT_ID(@Source_object)))+  
    N'.'+  
    QUOTENAME(OBJECT_NAME(OBJECT_ID(@Source_object)));
```



[Staging].[Currencies]



mitigations



sanitizing inputs

powershell

```
$email = "daniel@strd.co"; IF (SUSER_NAME()='sa') WAITFOR DELAY '00:00:01'; --"  
$email_escaped = $email -replace '"', '""'  
$sqlQuery = @"  
SELECT id, firstName, lastName  
FROM dbo.Users  
WHERE email='$email_escaped';  
"@
```



sql

```
SELECT id, firstName, lastName  
FROM dbo.Users  
WHERE email='daniel@strd.co'; IF (SUSER_NAME()='sa') WAITFOR DELAY '00:00:01'; --';
```



mitigations

using variables with Invoke-Sqlcmd?

powershell

```
$email = "daniel@strd.co"; IF (SUSER_NAME()='sa') WAITFOR DELAY '00:00:01'; --"
```

```
$sqlQuery = @"  
SELECT id, firstName, lastName  
FROM dbo.Users  
WHERE email= '$(email_addr)';  
"@
```

```
Invoke-Sqlcmd `  
...  
-Query $sqlQuery `  
-Variable "email_addr='$email'"
```



mitigations



using variables with Invoke-Sqlcmd?

powershell

```
$email = "daniel@strd.co"; IF (SUSER_NAME()='sa') WAITFOR DELAY '00:00:01'; --"

$sqlQuery = @"
SELECT id, firstName, lastName
FROM dbo.Users
WHERE email='$(email_addr)';
"@

Invoke-Sqlcmd `
...
-Query $sqlQuery `
-Variable "email_addr='$email'"
```



sql

```
SELECT id, firstName, lastName
FROM dbo.Users
WHERE email='daniel@strd.co'; IF (SUSER_NAME()='sa') WAITFOR DELAY '00:00:01'; --';
```



mitigations

using variables with Invoke-Sqlcmd?



powershell

```
$email = "daniel@strd.co"; IF (SUSER_NAME()='sa') WAITFOR DELAY '00:00:01'; --"

$sqlQuery = @"
SELECT id, firstName, lastName
FROM dbo.Users
WHERE email='$(email_addr)';
"@

Invoke-Sqlcmd `
...
-Query $sqlQuery `
-Variable "email_addr='$email'"
```



sql

```
SELECT id, firstName, lastName
FROM dbo.Users
WHERE email='daniel@strd.co'; IF (SUSER_NAME()='sa') WAITFOR DELAY '00:00:01'; --';
```



The -Variable switch is not parameterization



mitigations



parameterizing inputs in dbatools

powershell

```
$email = "daniel@strd.co"; IF (SUSER_NAME()='sa') WAITFOR DELAY '00:00:01'; --"

$params = @{
    email_addr = $email
}

$sqlQuery = @"
SELECT id, firstName, lastName
FROM dbo.Users
WHERE email=@email_addr;
"@

Invoke-DbaQuery `
...
-Query $sqlQuery `
-SqlParameters $params
```



mitigations



sanitizing unquoted inputs

powershell

```
$id = "123; DROP TABLE dbo.Students--"  
$id_typed = [int]$id  
$sqlQuery = @"  
SELECT id, firstName, lastName  
FROM dbo.Users  
WHERE id=$id_typed;  
"@
```



InvalidArgument: Cannot convert value "123; DROP TABLE dbo.Students--" to type "System.Int32". Error: "The input string '123; DROP TABLE dbo.Students--' was not in a correct format."



mitigations

sanitizing inputs

- There are theorized highly advanced exploits, where you could use specially crafted unicode strings with multi-byte characters.
- This is the only realistic option for languages where no parameterization can be done, like PowerShell.



mitigations



stored procedures with
restrictive permissions

- Only grant the application account EXECUTE permissions on stored procedures
- Do not grant SELECT, UPDATE, INSERT or DELETE

👉 This is not a sufficient mitigation on its own, but a good defense-in-depth strategy when combined with other mitigations.





mitigations

stored procedures with
restrictive permissions

- If you're brave enough, actually denying access to the system DMVs would prevent enumeration of tables, views, etc, but this could affect your application.

👉 This is not a sufficient mitigation on its own, but a good defense-in-depth strategy when combined with other mitigations.

sql

```
DENY SELECT ON SCHEMA::sys TO [databaseUser];  
  
DENY SELECT ON SCHEMA::INFORMATION_SCHEMA TO [databaseUser];
```



mitigations



disabling web server error messages

- Not showing error messages on your web page makes it much harder for attackers to use error-based injections

👉 This is not a sufficient mitigation on its own, but a good defense-in-depth strategy when combined with other mitigations.



mitigations



web application firewalls

- A web application firewall (WAF) can detect sql injection patterns and block the client from running repetitive exploratory queries.

👉 This is not a sufficient mitigation on its own, but a good defense-in-depth strategy when combined with other mitigations.



mitigations

rate limiting



- Timing-based attacks require *a lot* of http calls. Rate limiting your site may mitigate timing-based attacks, but is not a fully secure solution on its own.

👉 This is not a sufficient mitigation on its own, but a good defense-in-depth strategy when combined with other mitigations.



mitigations



sanitize connection strings

- Quote strings with apostrophes or double quotes
- Escape apostrophes/quotes accordingly

Not a very common vector, but relatively easy to mitigate.

```
Server=sql01.strd.co;Database=Prod_DB;User Id=daniel;Password=Pa$$w0rd!;Integrated Security=True;
```

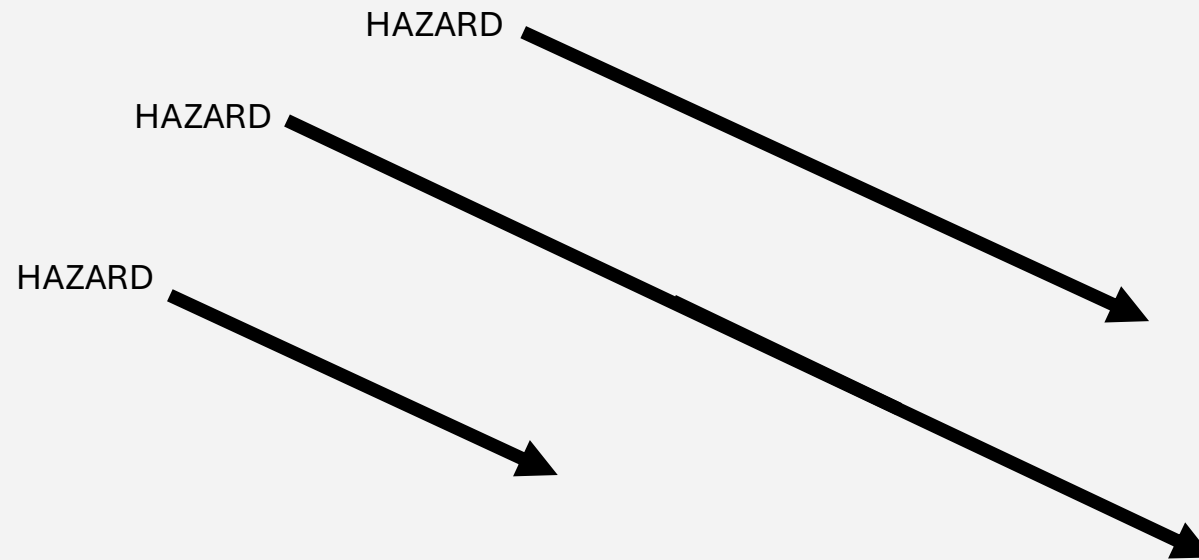


connection string

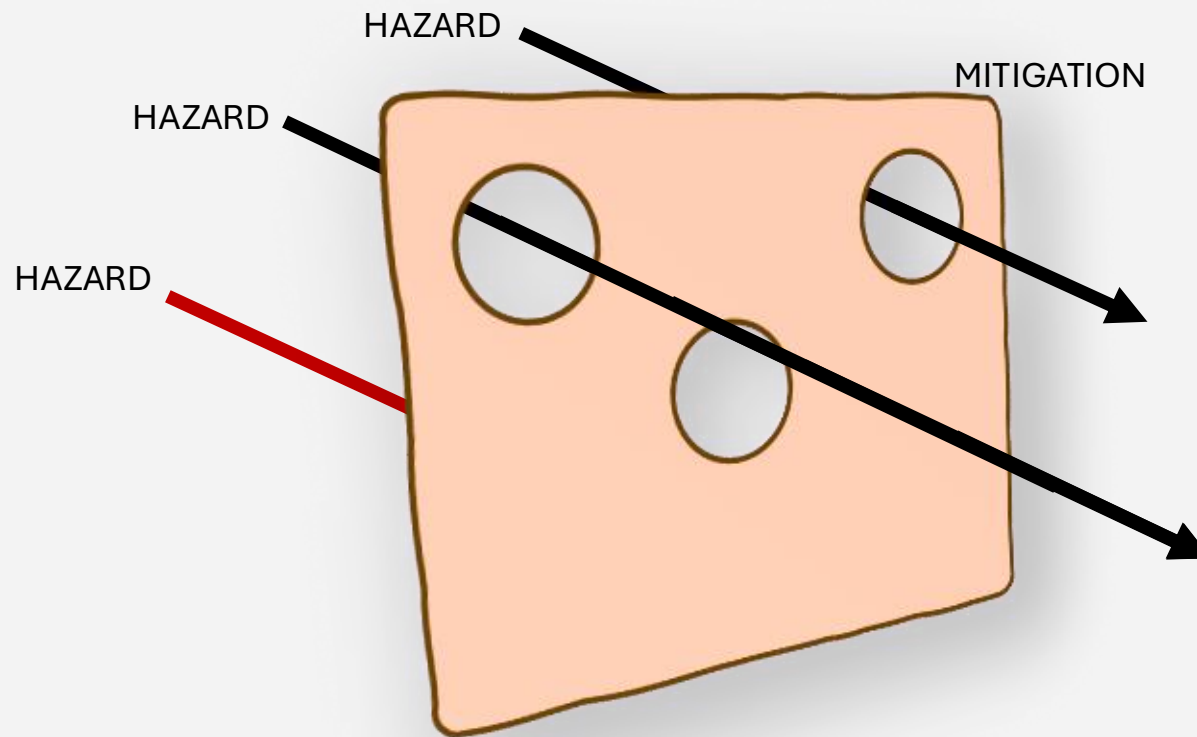
```
Server="sql01.strd.co";Database="Prod_DB";User Id="daniel";Password="Pa$$w0rd!;Integrated Security=True";
```



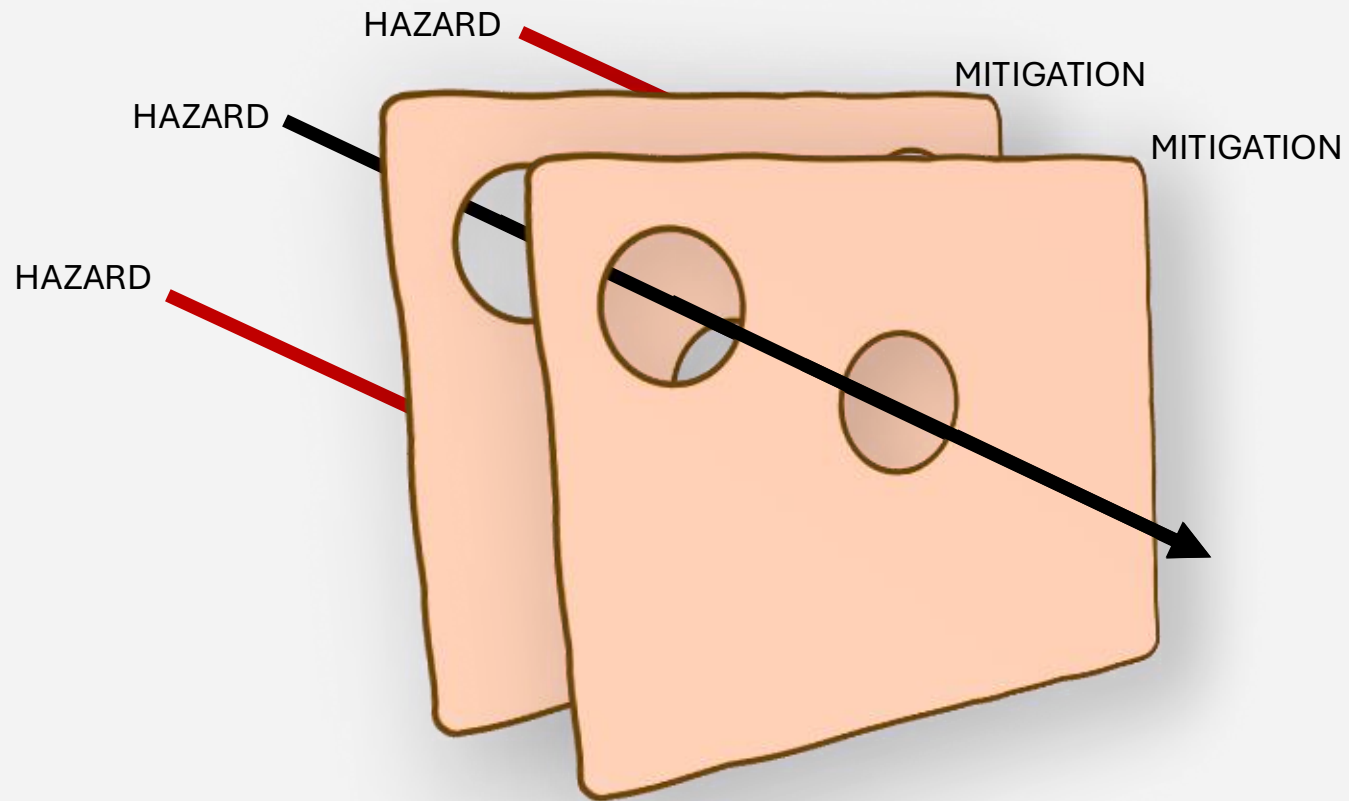
the "swiss cheese model"



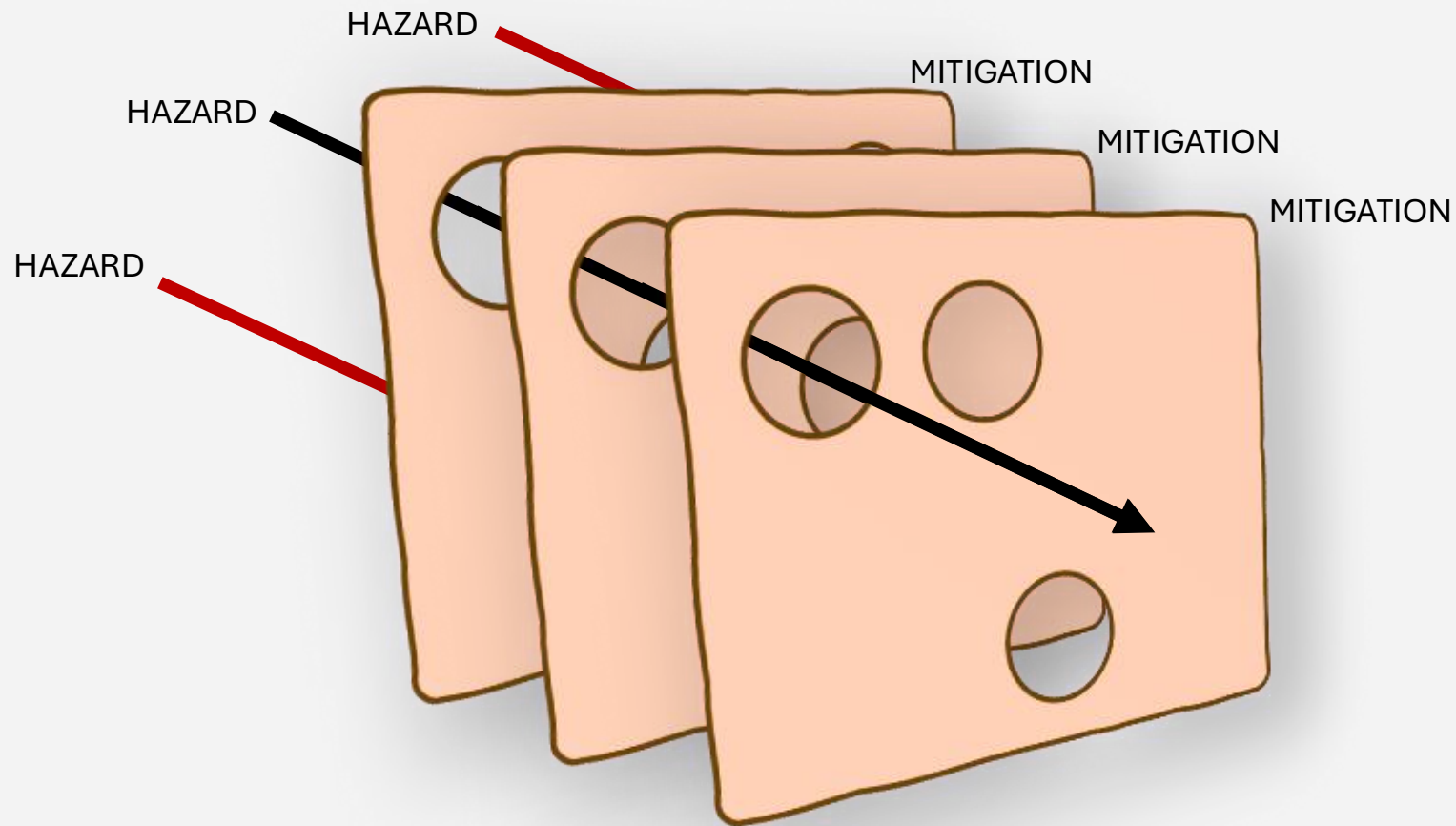
the "swiss cheese model"



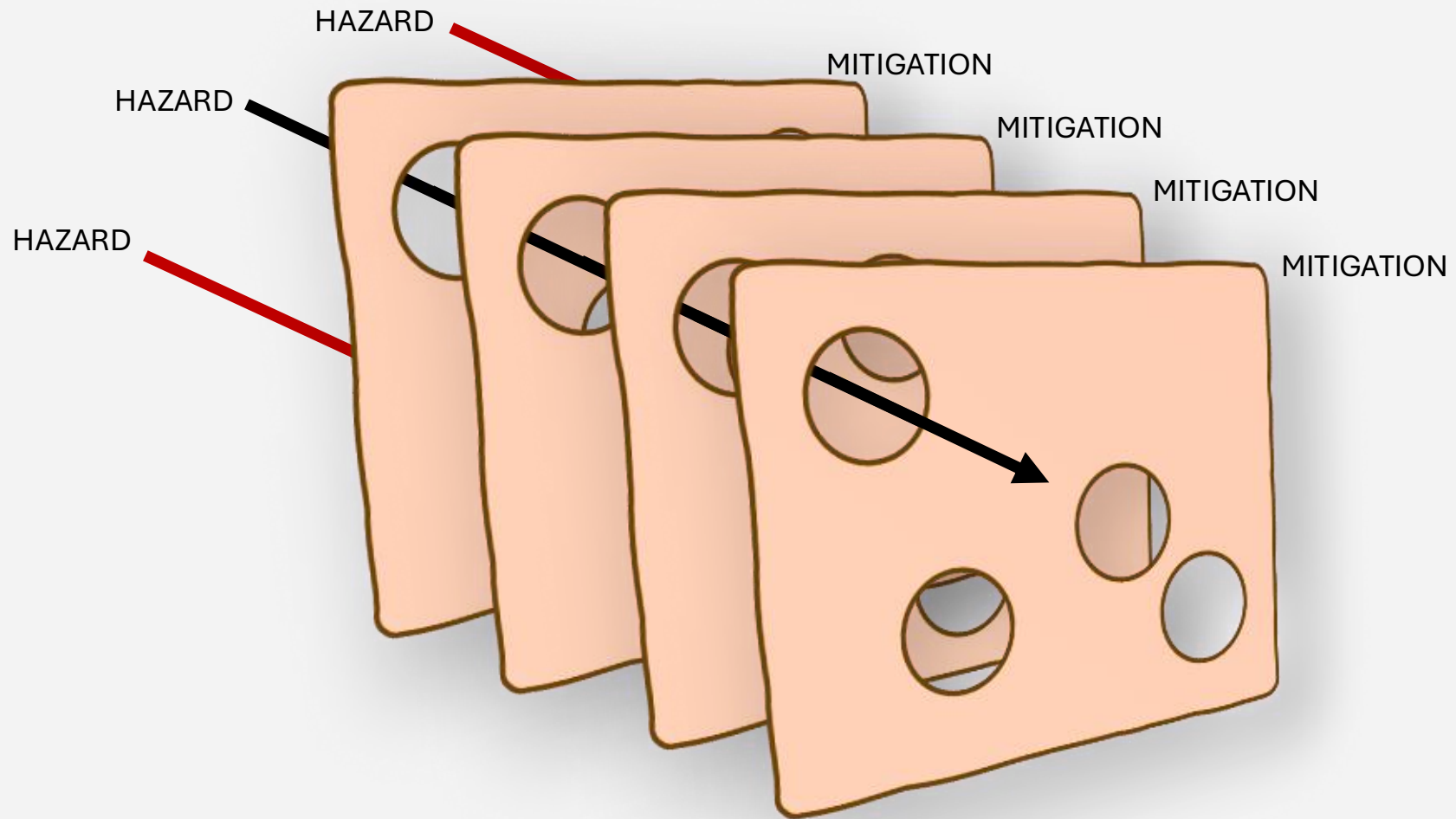
the "swiss cheese model"



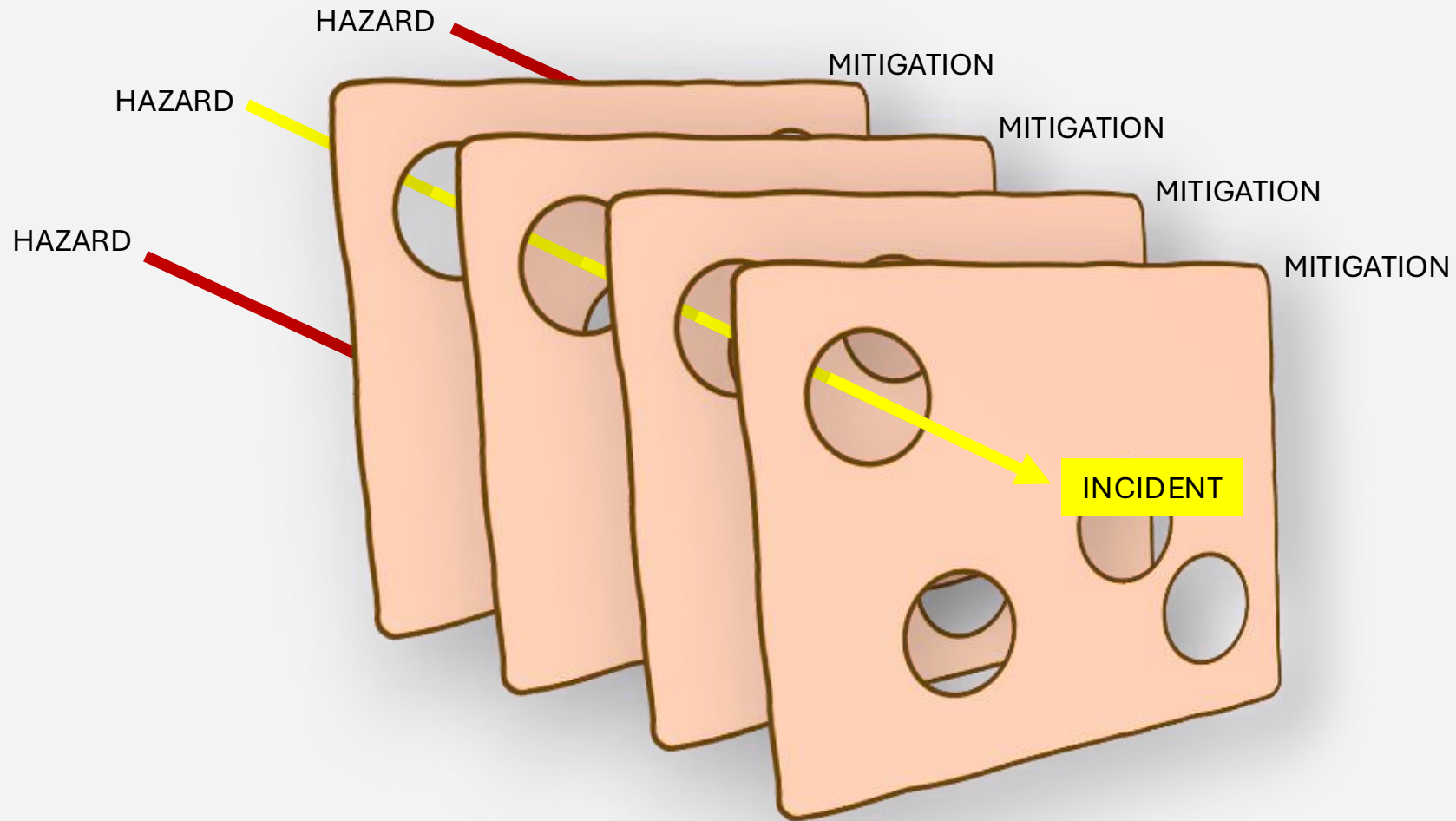
the "swiss cheese model"



the "swiss cheese model"



the "swiss cheese model"



mitigations

✅ = Complete mitigation
✓ = Partial mitigation

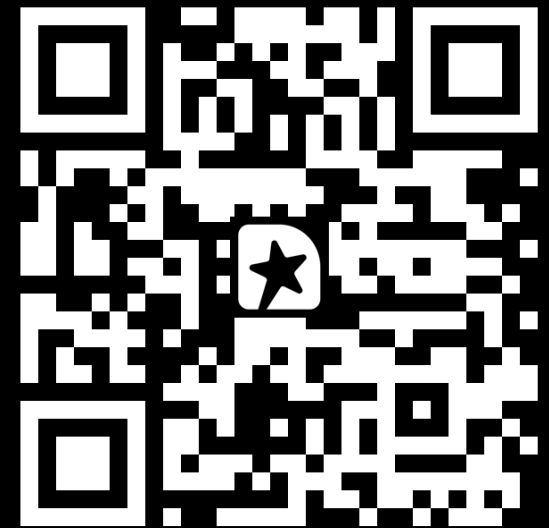
	Parameterized inputs	ORM	QUOTENAME	Sanitizing inputs	Restricted permissions	Disabling error messages	Web Application Firewall	Rate limiting
Out-of-band injections	✅	✅		✅	✅			
Error-based injections	✅	✅		✅	✓	✅	✓	✓
Blind injections	✅	✅		✅	✓		✓	✓
Escaping dynamic SQL			✅	✅	✓			
Connection string injection	✅ *			✅				

← defense in depth →

'-- questions?



feedback



'-- thank you



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slides and demos

github.com/sqlsunday/presentations

github.com/sqlsunday/injection-demo